

RODJEAN VERZOSA

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SUMMARY

Experienced Software Engineer with a strong background in full-stack web development and cross-platform mobile applications (iOS & Android). Skilled in designing and delivering robust, scalable, and user-friendly solutions that drive operational efficiency and enhance user experience. Adept at leveraging machine learning and data-driven models to inform product features and optimize performance. Proven ability to architect complex systems, integrate cloud and IoT technologies, and translate business requirements into technical solutions.

WORK EXPERIENCE

Senior Software Engineer, iKomyutPH (Fullstack)

Aug 2025 - Present

- Architected and delivered the company's production-ready bus ticketing system, deployed on onboard ticketing devices and built for high availability and seamless synchronization.
- Driving end-to-end development of the broader fleet intelligence platform, encompassing GPS tracking, onboard camera integration, driver behavior analytics, real-time event streaming, and operational insights.
- Leading the creation of a real-time command and monitoring dashboard for fleet operators, providing actionable visibility into bus locations, trip performance, device status, and driver behavior events.
- Spearheading backend service design, API development, and scalable IoT data pipelines handling continuous telemetry from ticketing devices, GPS modules, and edge cameras.
- Overseeing full-stack system architecture across mobile (Ionic), web (Next.js), and backend (NestJS), ensuring modularity, maintainability, and alignment with long-term product growth.
- Managing cloud infrastructure across GCP and AWS, including compute services, messaging queues, real-time communication systems, CI/CD pipelines, and security hardening.
- Designing and optimizing database structures in MongoDB, supporting high-volume transactions, geospatial queries, and telemetry storage.
- Implementing robust device-cloud communication strategies using WebSockets, and REST APIs to support real-time updates and fault tolerance.
- Collaborating closely with transport operators, drivers, and stakeholders to validate features, gather field insights, conduct pilot tests, and guide product direction based on operational realities.
- Driving cross-functional alignment by translating business requirements into scalable engineering solutions, setting technical standards, and mentoring contributors.
- Coordinating roadmap planning, feature prioritization, and deployment strategies to ensure smooth rollouts, backward compatibility, and high system uptime.

Tech Stack:

- Frontend: Ionic, Next.js, Backend: Node.js ecosystem, NestJS, Database: MongoDB, Cloud: Google Cloud Platform (GCP), Other: WebSockets, REST APIs, IoT device integration

- Designed and developed a full-featured, cross-platform ride-hailing mobile application using Capacitor, supporting both Android and iOS. Focused on performance, responsiveness, and native like user experience, while enabling fast development cycles with a single codebase.
- Refactored and migrated complex .NET legacy logic to Node.js, ensuring feature parity, optimizing performance, and implementing modern JavaScript/TypeScript patterns while reducing technical debt.
- Implemented real-time driver–rider matching, GPS tracking, and route optimization using Google Maps APIs. Integrated turn-by-turn navigation, ETA calculations, and live geolocation to ensure seamless coordination between riders and drivers.
- Integrated 2C2P for secure payment processing, supporting multiple payment methods including mobile wallets. Engineered dynamic fare calculation, trip invoicing, and refund handling, ensuring compliance with payment regulations and improving user trust.
- Built robust, scalable backend APIs using Node.js with Nest.js, adhering to RESTful best practices. Deployed services on AWS using containerized architectures to ensure high availability, fault tolerance, and horizontal scalability.
- Developed and integrated WebSocket-based real-time communication channels between riders and drivers. Enabled low-latency updates for ride status, driver arrival, trip progress, and cancellations, improving user experience and operational reliability.
- Developed an internal admin dashboard using Next.js for end-to-end management of users, drivers, trip logs, payments, and analytics. Included role-based access control, audit trails, and real-time dashboards for operational oversight.
- Deployed mobile applications to both Google Play Store and Apple App Store, managing app signing, release notes, versioning strategies, phased rollouts, and compliance with each platform's guidelines.
- Optimized Google Maps API usage to reduce monthly billing costs by over 40%. Techniques included intelligent caching, rate-limiting API calls, deferring expensive queries, and leveraging static or lite map modes where appropriate.
- Implemented push notifications and real-time updates using Firebase Cloud Messaging (FCM) and WebSockets, ensuring timely alerts for ride confirmations, driver arrivals, and system events without compromising battery efficiency.
- Continuously monitored application health and performance using AWS CloudWatch. Created alerting dashboards, triaged bugs, resolved crashes, and rolled out hotfixes quickly to maintain high uptime and user satisfaction.
- Maintained and refactored legacy codebases across both mobile and backend services. Migrated obsolete logic to modular, testable components, removed redundant code, reduced technical debt, and improved long-term maintainability.
- Identified and mitigated a security breach involving an improperly configured AWS S3 bucket. Resolved the issue by enforcing principle of least privilege with stricter IAM roles, adding server-side encryption, lifecycle rules, and tighter bucket policies to secure sensitive assets.
- Collected and analyzed user feedback through in-app analytics, support tickets, and field reports. Prioritized feature enhancements and UX improvements that reduced rider drop-offs and increased repeat usage and retention.
- Mentored junior developers through pair programming, code reviews, and architectural discussions. Helped standardize development practices, improve onboarding time, and instill clean coding habits across the team.
- Integrated Promotexter SMS OTP into NestJS, delivering secure time-bound verification codes with expiry, retry limits, and MongoDB-backed audit logging.

Software Engineer, Kakao Story (Machine Learning)

May 2024 - Dec 2024

- Developed an autonomous Baccarat Bot implementing a modified betting system with customizable parameters, enabling users to tailor betting strategies dynamically to optimize performance under varying game conditions. Integrated simulation testing to evaluate system efficacy and adjust strategy parameters.
- Created an autonomous Blackjack Bot featuring a modified betting system alongside an advanced card counting algorithm (e.g., Hi-Lo system), allowing the bot to adapt betting and playing strategies in real-time to maximize winning potential. Implemented decision-making logic based on statistical probabilities and risk management principles.
- Developed user interfaces for monitoring bot performance and customizing strategy parameters, using Python Tkbootstrap, enhancing user control and transparency.
- Developed an autonomous Poker AI integrating with online poker APIs for real-time data ingestion. Leveraged machine learning models using Scikit-learn and TensorFlow, including reinforcement learning, to predict opponent behavior and optimize strategy. Designed and deployed a Flask-based web API for managing extensive training data, with secure and efficient data handling. Deployed a scalable web application on Google Cloud Platform, utilizing App Engine, Cloud Storage, and Cloud SQL, supported by automated CI/CD pipelines. Built real-time data pipelines to preprocess live game data, enhancing model accuracy and adaptive gameplay.

Data Analyst, Accenture Inc.

Feb 2018 - Mar 2024

- I am skilled in evaluating the accuracy and quality of data in medical systems, focusing on medical records and managing prior authorizations. My expertise includes calculating coinsurance, deductibles, and copays, adhering to HIPAA regulations for confidentiality. I proficiently handle end-to-end claims processing, verify member status, adjust claims, and route them accurately to relevant departments.

EDUCATION

Bachelor of Science in Information Technology

2013 - 2017

Colegio de Montalban

- Thesis on "Cardio Monitoring using Arduino".
- Projects on "Casino Card Dealer"

ADDITIONAL INFORMATION

- **Technical Skills:** Python, JavaScript, TypeScript, Capacitor, iOS, Android, Next.js, React.js, Node.js, Express, Flask, .NET Core, Scikit-learn, TensorFlow, Reinforcement Learning, Google Cloud Platform, AWS, WebSocket, MongoDB, RESTful API design, secure authentication, Git, Arduino
- **Certifications:** JavaScript Algorithms and Data Structures, Responsive Web Design, Python essentials